

1916

Sharing electrons

American chemist Gilbert Lewis suggested that when atoms bond together to form molecules they share their outer electrons. The idea was developed further by another American chemist, Irving Langmuir, in 1919, and is known as the Lewis-Langmuir Theory.

The story of the atom
See pages 168–169

Chemist Ernest Rutherford changed atoms of nitrogen into oxygen in 1917 by firing particles at the atoms' nuclei—a process called transmutation.

Everything in its place

In 1917, American zoologist Joseph Grinnell introduced the idea that every creature has its own place or role in its habitat (home), known as an ecological niche.

Dung beetles eat and bury dung, increasing the amount of nutrients in the soil and making a habitat more liveable for other creatures.

Airship frame measured 643 ft (196 m) long



R34 airship crosses the Atlantic, 1919

1916

Milky Way location

American astronomer Harlow Shapley established that our solar system is not at the center of the Milky Way as many previously thought, but, in fact, thousands of light years away. Shapley came to this conclusion after he studied clusters of distant stars and found that they formed a halo around the center of the Milky Way.

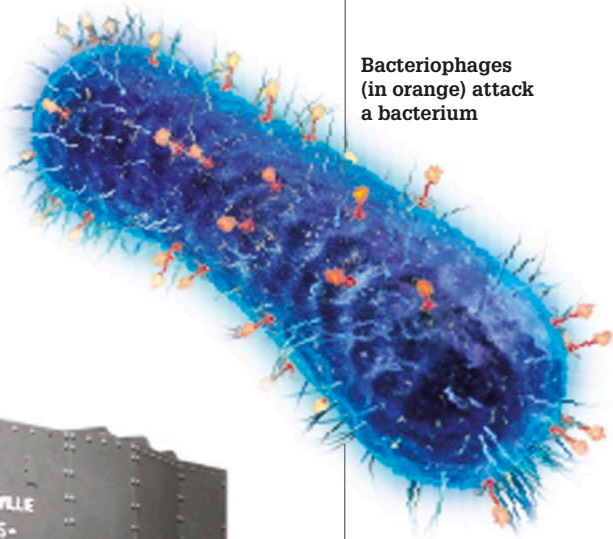
1919

First aerial crossings of the Atlantic

Three different types of aircraft successfully crossed the Atlantic in 1919: the first airship (the R34), the first flying boat (an NC-4), and the first nonstop flight made in a Vickers Vimy plane by British aviators John Alcock and Arthur Brown.

1920

Bacteriophages
(in orange) attack a bacterium

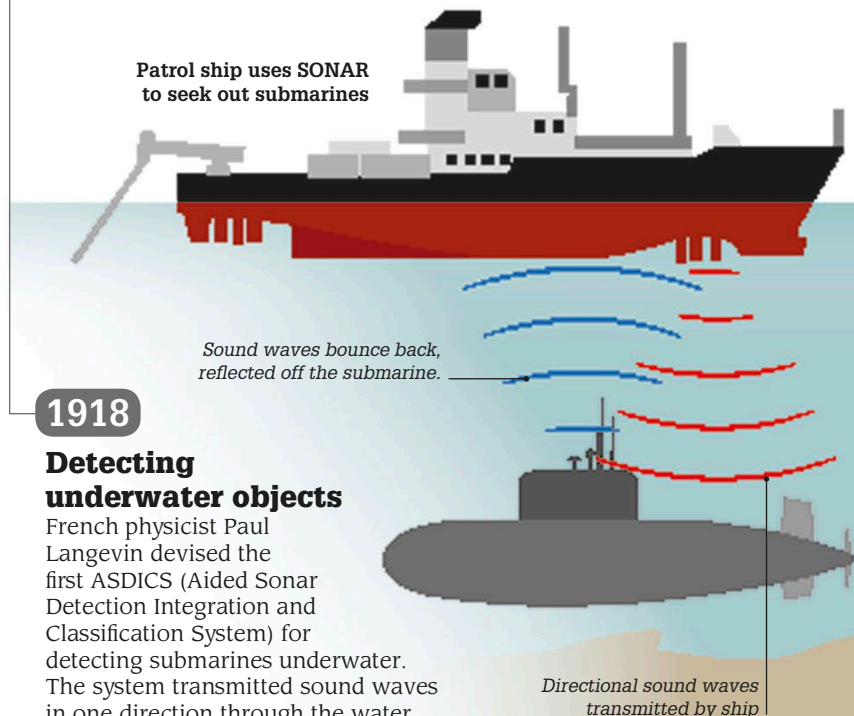


1917

Bacteria eaters

While investigating the bacteria that cause the disease dysentery, French-Canadian biologist Félix d'Hérelle discovered and named viruses that attack and destroy bacteria. He called them bacteriophages, meaning bacteria eaters.

Patrol ship uses SONAR to seek out submarines



Sound waves bounce back, reflected off the submarine.

1918

Detecting underwater objects

French physicist Paul Langevin devised the first ASDICS (Aided Sonar Detection Integration and Classification System) for detecting submarines underwater. The system transmitted sound waves in one direction through the water, and measured distances by how long it took signals to reflect back. The ASDICS was the forerunner of modern SONAR (Sound Navigation And Ranging) technology.

Directional sound waves transmitted by ship